

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2-CONCEPT MAPPING

Course Code:

CSA12

CO Assessed:

CO1

Precision (BL4)

Weightage::

35%

29

Course Title:

Assessment:

Computer Architecture

Concept Mapping

Total Marks:

100

CO1 Statement:

Student Name:

Section / Batch:

Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking

Shen Ivin
Clement

2023

2027

Reg. No.:

Date:

192371023 20-07-26

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Instructions to Students

Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.

Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.

Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases. Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.

Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

1. A software company is upgrading its office computers and must decide whether to use ARM Cortex, Intel Core, or AMD Ryzen processors for different workloads. Construct a concept map linking these modern architectures with their instruction execution, cache usage, processing capability, and application areas. Show how their architectural features influence performance and energy efficiency. How do modern processor architectures differ in meeting the requirements of mobile, desktop, and high-performance computing systems?
2. A computer technician is troubleshooting a system that stores machine instructions in memory but displays memory addresses in a debugger. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU interpretation, and data conversion. Show how instructions are represented, stored, and interpreted during execution. How do binary and hexadecimal number systems simplify memory organization, instruction representation, and debugging?
3. A manufacturing company is upgrading its automated inspection system and notices delays when multiple sensors and storage devices communicate with the processor simultaneously. Construct a concept map linking CPU, Memory, I/O Devices, Single-bus, Multi-bus, and Hierarchical Bus structures, showing how data and control signals are transferred. Illustrate the effect of each bus organization on communication efficiency. How do different bus structures influence system performance and reduce communication bottlenecks?
4. A programmer is developing an industrial automation application and must choose the most suitable instruction format and addressing mode for faster execution and efficient memory access. Construct a concept map linking Instruction Formats, Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control Instructions. Show how each addressing mode affects execution speed and memory access. How do instruction formats and addressing modes influence program execution and processor efficiency?
5. An embedded systems engineer is comparing the 8085 and 8086 microprocessors for a factory monitoring

system that **requires** efficient instruction execution and memory organization. Construct a concept map linking their **instruction sets**, **register** organization, addressing modes, and memory organization, including memory **segmentation** in 8086. Show **how these** features affect program execution and system performance. How do the **architectural** differences **between** the 8085 and 8086 influence execution efficiency and memory management?
Code of Conduct

*I Sheo him Clement **192371023** (Name / **Reg** No) certify that **this submission is my original work and that I have adhered to the guidelines specified for this assessment.** I understand that any violation of **academic integrity** rules will **result** in disciplinary action.*

Signature of the Student:

ВИНЫ

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
		(5 Marks)	(4 Marks)		
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Establishes clear, meaningful, and accurate		
Linkages	25%	relationships between concepts			
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Integrates multiple studies effectively		

**Integration of
Literature**

20%

to show
connections and
comparisons

Identifies **most** key
concepts with **minor**
omissions

Shows **correct**
relationships with **minor**
gaps

Mostly organized with
minor structural issues

Shows **some**
integration of
literature

(2-3 Marks)

Identifies limited concepts;
some are irrelevant **or**
missing **matig**

Relationships are
weak, unclear, or
partially incorrect

Some organization but
lacks clear hierarchy

Limited integration;
mostly isolated
concepts

(0-1 Mark)

Fails to identify
essential concepts

				No meaningful relationships established
				No logical organization or structure
				No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts
Marks Evaluation:				
				Poorly presented and hard to understand
Q.	Concept	Organization	Integration of Literature	Clarity (10)
No	Linkages (25)	(20)	(20)	Total (out of 100)
	Identification (25)			

	/4		/4		/4		/4		14
2	/4		/4		/4		/4		/ 4
3	/4		/4		/4		/4		/4
4	/4		/4		/4		/4		/4
5	/4		/4		/4		/4		/4
	/25		/25		/20		/20		/10
									/ 100

Course Coordinator

Faculty Incharge

Processor Architectures: ARM Cortex Vs Intel Core's AMD Ryzen Prousson

Architecture

ARM Cortex
(RISC)

Inkt come

AMD Ryzen

CCISC /
Hybrid

(CISL X86)

Simple,
fined

Complexe,
Variable

length, micro

length, Single
Cycle

Complex
x86-64 SMT
threading

Small low-power

horge
multilend

21122

Lau- clock

high penfl watt

Mobile TOT

embedded

4/12/13

4

High clock

Strong

Single No

thread

Desktops,

Laptops

hait 23

Chiplet design

high Count
high
throughput

Gaming,
konues,

HOC

Performance & Energy
Efficiency

2.

Binary & Heno Decimal
Number System

Number Systems in
computing

Binary

System (
Brace 2)

Henadicional
System]
(Bore 16)

Instruction Rep

Memory Storage:
Compact form.

opcode as Os/01

bit pultum
in

Registus /cells

It bits = 1 hex

digits

Address &
Debugger
Display: eg:
Ox/A3F

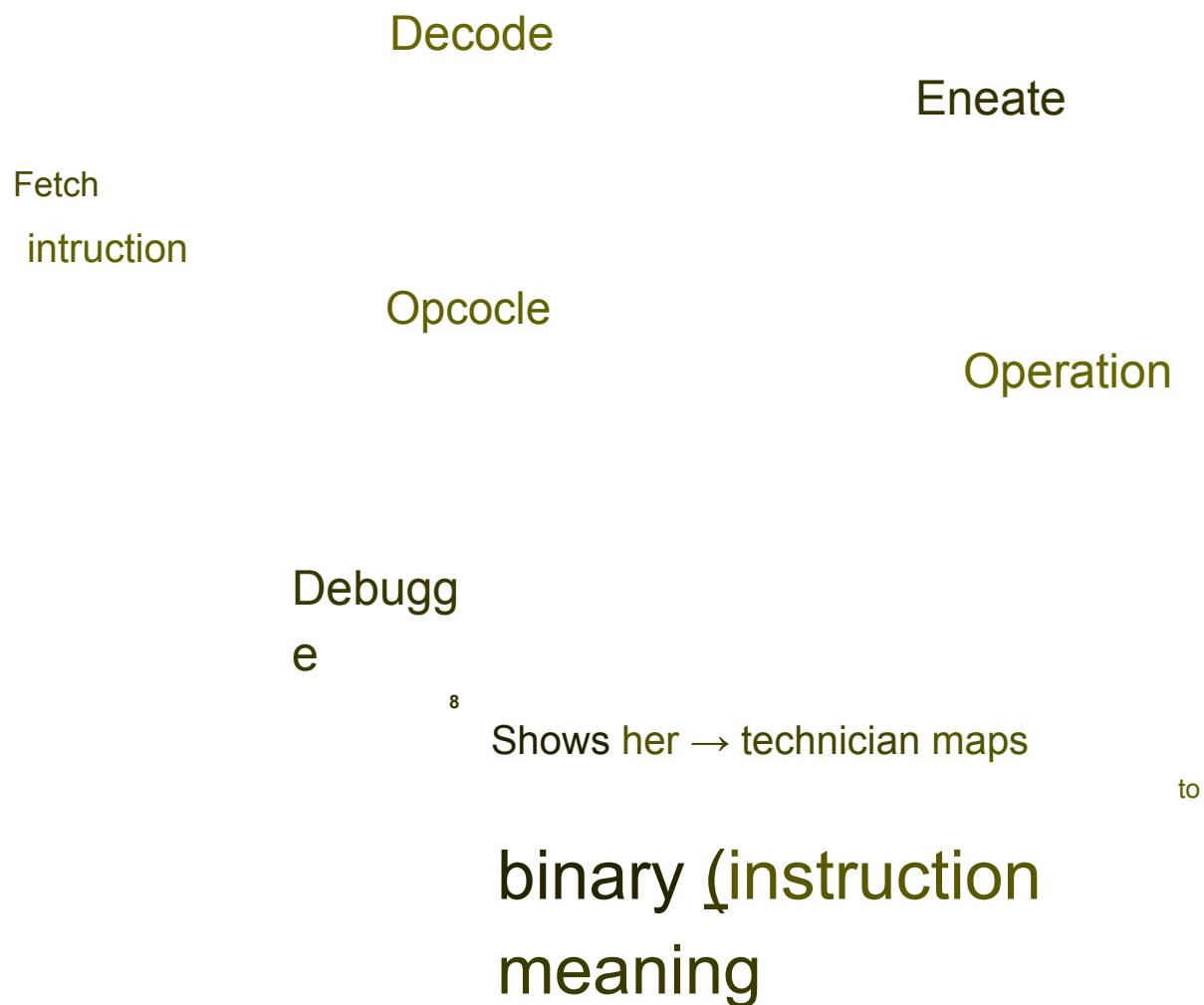
3.

Dana Conversion

(group leexpand 4 bits <—> 1 hen
cligit)

CRV interprctation

Contooch chuit decodes
binary Opcode



Bus Structures: Single Bus, Multi Bus, Hierarchical Bus

CPU

Single

Bus

Structure

One shared

bus for
data/address

Control

Memory

- I/O Devices

But

Organization

MultiBus

structure

Parallel

buses

for different

Components

Cache
(Storage),

Parallel

Bus

Stanchion

levels: local
bus,

System bus, be
pansion
bus

Only 1

denic
e

Simultaneous

us

transfers at
a time

Bridges manage

transfer

s

Seaweed
d

traffic -7
minis

→ BOTTLE NECK

contention

System

Performance

And

Communication
Bottle Necks

Instruction Formats And Addressing Modes

Instruction format

Immediate

Cop

W

Coplo

de

ode +

Operand
Files)

[Direct

Register Inclinet

Imeloncel

Addlarning Addheking Adddering
Addduring Adddering

FASTEST : nos entra

(Moderate : 1

memory access

(Immediak,

Regi
ster)

Backa

Transfer

instructions

CLOAD/'STORE) -

Register

Direct, Indirect

ашень

SLOWED:

multiple

memory acces

C. Dissect,

inclined)"

memory

accersis

Cindiret

Anithanatic

Control Intructions

Trutructions

(ADD, SUB,

A-IUL)

CJump

(Brunch

Dirut,

Inclened

Immediate,
Register

Execution feed and
Memory Access
Efficiency

5;

5 .) : Micso Processor

Comparisons: 8085 vs 8086

Microproceson
Comparison

8085

(8-bit)

8-bit instructions

Simple
oprodes

6x8 bit GP cregisters

(B. C. DIE, HIL) +

Accumulator

Dilouet, Register,
Immediate,
Inclined

Indirect

16 **but** addues bus,

64KB Max, No

Segmentation

Sutable for
Simple,
low cost,
monitoring

for les

8086

(16-bit)

16-bit

16-bit instructions

Archer, Dower

full,

OpCodes

16-bit Conventions (AX, BX, CX, DX) + Segment

Registers (CS, DS, SS, ES)

Register, Immediate, Discrete, Indirect, Based

SEGMENTED

D

SEGMENTED memory Segment offset, 1MB

Space

Suitable for complex **factory monitoring**

large data,
multitasking

es